

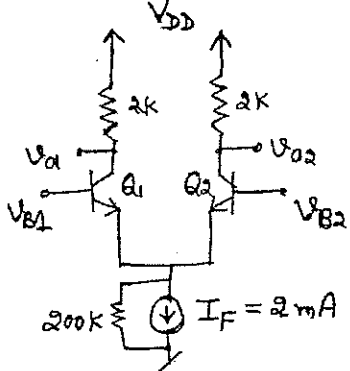
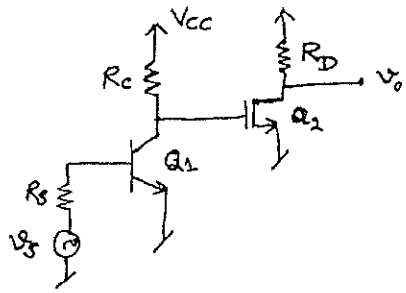
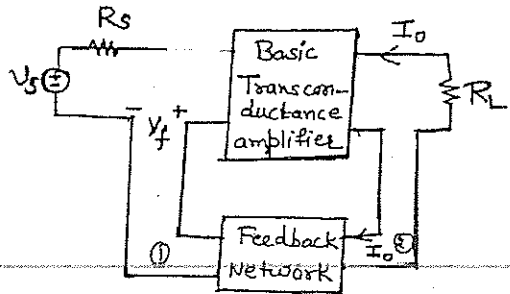
Course Code: ECECC-304

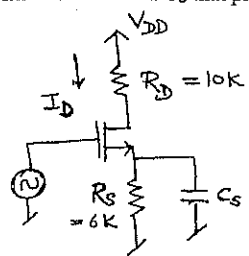
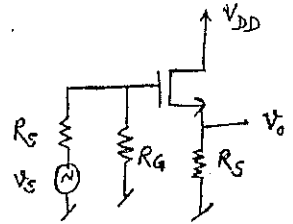
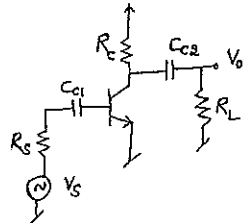
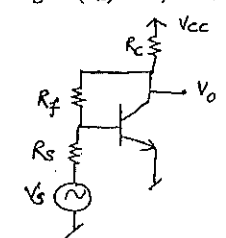
Course Title: Microelectronics Circuits and Applications

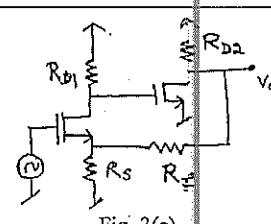
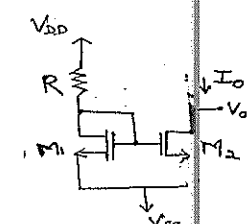
Time: 03:00 Hrs.

Max. Marks: 50

Note: Attempt all the five questions. All questions carry equal marks. Missing data/information if any will be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
Q1	Attempt any 2 parts of the following		
1(a)	i) For the Differential Amplifier shown in Fig. 1(a), find A_D and A_{CM}.  Fig. 1(a) ii) Define CMRR for differential amplifier and calculate the same for the circuit shown in Fig. 1(a).	(3+2)	CO1
1(b)	For the circuit shown in Fig. 1(b): i) Draw small signal equivalent circuit. ii) Find overall gain.  Fig. 1(b)	(1+4)	CO1
1(c)	i) A series-series feedback circuit representable by Fig. 1(c) and using ideal transconductance amplifier operates with $V_S = 100\text{mV}$, $V_f = 95\text{mV}$ and $I_O = 10\text{mA}$. What are the corresponding values of A and β. Include the unit in each answer.  Fig. 1(c) ii) Find R_{if} in terms of R_i, A and β for series-series topology.	(3+2)	CO3

Q2	Attempt any 2 parts of the following		
2(a)	For the amplifier shown in Fig. 2(a), is biased to operate at $I_D = 1\text{mA}$, $g_m = 1\text{mA/V}$, $R_D = 10\text{k}\Omega$, $R_S = 6\text{k}\Omega$. Neglecting r_o, find the values of C_S that places f_L at 100Hz.  Fig. 2(a)	(5)	CO2
2(b)	For the circuit shown in Fig. 2(b) draw small signal high frequency equivalent circuit and find higher cut of frequency (f_H) by assuming the dominant pole approximation is valid.  Fig. 2(b)	(5)	CO2
2(c)	For the common emitter amplifier shown in Fig. 2(c), find lower cut of frequency (f_L).  Fig. 2(c)	(5)	CO2
Q3	Attempt any 2 parts of the following		
3(a)	Draw the feedback amplifier in signal flow graph form/ block diagram and find gain with feedback (A_f), assuming that negative feedback is applied. Draw the series-shunt topology and find its R_{if} and R_{of} in terms of A and β.	(5)	CO3
3(b)	For the circuit shown in Fig. 3(b), with following parameters: $R_C = 4\text{k}\Omega$, $R_F = 40\text{k}\Omega$, $R_S = 10\text{k}\Omega$, $r_\pi = 1.1\text{k}\Omega$, transistor current gain (h_{fe}) = 50, find A_f, R_{if}, and R_{of}.  Fig. 3(b)	(5)	CO3
3(c)	For the series-shunt amplifier shown in Fig. 3(c), find A_f, R_{if}, and R_{of}.	(5)	CO3

	 Fig. 3(c)		
Q4	Attempt any 2 parts of the following		
4(a)	Draw the circuit of MOSFET based cascode current mirror and derive the expressions for its output resistance. Also list ideal characteristics of a current mirror.	(5)	CO4
4(b)	i) Draw the circuit of class B push-pull amplifier and derive its relation of conversion (overall) efficiency. ii) Class B push-pull amplifier with $V_{CC} = 25\text{V}$ drives a 8Ω load. Determine the maximum input power, output power, efficiency and power dissipated by each transistor.	(2+3)	CO4
4(c)	Design a MOSFET based current mirror to obtain an output current of $50\mu\text{A}$ (Fig. 4(c)). $V_{DD} = 1.8\text{V}$, $I_{ref} = 50\mu\text{A}$, find R, if M_1 and M_2 are matched with $L = 0.5\mu\text{m}$, $W = 5\mu\text{m}$, $V_{th} = 0.5\text{V}$, $K_n = 250\mu\text{A/V}^2$.  Fig. 4(c)	(5)	CO4
Q5	Attempt any 2 parts of the following		
5(a)	State the Barkhausen Criteria. Draw MOSFET based Wein-bridge oscillator and derive its expressions for condition of sustained oscillation and frequency of oscillation.	(5)	CO5
5(b)	Draw and explain working of first order low pass and high pass active filters using Op-Amp having 10kHz cut-off frequency and gain = 10V/V.	(5)	CO5
5(c)	Draw the circuit of two-integrator loop topology based KHN filter and derive its expressions for high-pass, band-pass and low-pass filters.	(5)	CO5